



Hydrocarbon Migration & Source-Rock Notes

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Audience: technical

I) Hydrocarbons That Had To Form In Place (p.1)

A) HC Source Rocks That Are Now Reservoir Rocks (Low Permeability)

1. Bakken •
2. Barnett •
3. Brown Dense •
4. Cherokee
5. Eagleford •
6. Fayetteville
7. Haynesville
8. Kimmeridge
9. Lewis
10. Mancos •
11. Marcellus
12. Monterey •
13. Mowry
14. Niobrara •
15. Pearsall
16. Sunniland •
17. Tuscaloosa •
18. Utica–Point Pleasant •
19. Vaca Muerta •
20. Woodford •
21. Wolfcamp Shale, Bonespring & Spraberry

(• marks appear next to roughly half the entries in the original — likely flagging entries used elsewhere, e.g. list II below.)

3) Isolated Sand Lenses — Completely Encased By Shales

2. San Juan Basin - Gallup SS, Tropic SS, Elvado SS
28. DJ Basin, Powder River Basin, Muddy SS
24. Kansas & Eastern Oklahoma - Cherokee SS
3. Powder River Basin - Muddy SS
5. Delaware Basin - Atoka LS, Morrow SS
6. Michigan - Shinarump SS
7. Colorado & Wyoming - Fractured Niobrara Shale

- 8) Pennsylvania, West Virginia, Ohio
9) Michigan - Silurian Reefs

- 4 25. Delaware Basin (Bone Springs [?]), Atoka LS & Morrow SS
- 5 26. Michigan, Shoestring SS
- 6 27. Colorado, Wyoming, Fractured Niobrara Shale
- 7 28. Pennsylvania, West VA & Ohio, Many SS
- 8 29. Michigan, Silurian Reefs

(numbering in the original runs 1–9 but two items are bracketed together under one number — transcribed here as closely as legible.)

l) continued — narrative (p.2–3)

Recent advances in the technology of horizontal drilling coupled with multistage hydraulic fracture stimulation has resulted in major oil and gas production increases in the United States. This technology is also being applied in other countries such as Canada, China and Argentina but on a much smaller scale. The wells are being drilled in what has become known as "unconventional shales." These shales are really a mix of shales (clay size particles), siltstones (silt size particles), sandstones (sand size particles) and carbonates. These formations have been drilled through by hundreds of thousands of wells in the past and were the large majority of the known and suspected source rocks that sourced the oil and gas of most of the conventional oil and gas production fields in these areas.

The pores in these source rocks are so small (1nm to 100nm, .001 to 0.1mm) and the flow paths are so tortuous that these rocks have permeabilities in the same (10–100 nanodarcy) range as most seals on overpressured reservoirs. ~~[struck: "on unconventional/conventional reservoirs"]~~ This is why average recoveries are so low — 5 to 10% in gas shales and 2 to 7% in oil shales — even with horizontal wells of 1 to 2 miles in length with multistage fracture treatments using in excess of 20 million pounds of sand.

This proves that these hydrocarbons had to originate in-situ in these source rocks and could not have migrated into these rocks. It also makes it highly unlikely that any primary migration of hydrocarbons out of these rocks has occurred ~~[struck: "especially"]~~ at current permeabilities. Migration had to take place prior to large-scale compaction and diagenesis taking place, when they had much larger permeabilities (100 to 1000 md or more). The only way this could occur is with very rapid rates of deposition that would allow the source rocks to reach the depths and temperatures required for major hydrocarbon generation prior to and during compaction.

Primary oil migration could then occur while large volumes of pore water was being expelled by the compaction process. The source rocks and surrounding formations would still have significant permeabilities that would allow both vertical and long-distance secondary migration into traps formed by the time of migration. Traps formed after migration occurred would therefore be barren of hydrocarbons. Many examples have been tested. See section VII.E.

Mississippian Barnett Shale (p.4–6)

[marg. top of p.4: "I) A) 2 & II) 1" — cross-reference to the Barnett entries in lists I)A and II) above/below]

The Mississippian Barnett Shale is an unconventional gas play in the Bend Arch – Fort Worth Basin of Texas. It includes parts of the Dallas–Fort Worth metroplex area. Exploration and production of oil and gas began in the early 1900's. Prior to 1998 [marg: "and amounted to approximately 2 billion bbl oil and 7 TCF of gas"] the production was mostly from conventional reservoirs ranging in age from Ordovician to Permian. Detailed biomarker and light hydrocarbon geochemistry confirm that the marine Mississippian Barnett Shale is the primary source rock for petroleum in the Fort Worth Basin (Hill, Jarvie & others 2007). Gases in the Fort Worth Basin are thermogenic in origin and appear to be cogenerated with oils from the Barnett Shale, although some gas may also originate by oil cracking. The gases appear to be stratigraphically segregated — younger reservoirs contain less mature gas and older reservoirs contain more mature gas, except in the Pennsylvanian Bend Group (Hill, Jarvie, others 2007). This segregation of gases indicates multiple or continuous migration times. [marg: "It is highly unlikely that this migration took place before compaction and diagenesis reduced the permeability of the B.S. down to the current nanodarcy range."]

Plots of gas chromatograph results for oils derived from various portions of the basin suggest a common source with differences in maturation accounting for the specific range of values. Oils derived from the Barnett from many different locations and with widely differing levels of maturity show highly similar biomarker signatures (Figure 5-5, *Petroleum Frontiers*) (Jarvie et al. 2001), indicating that fingerprinting has established the Barnett as the source for oils produced from the following intervals:

- 30. Ordovician: Ellenberger, Viola
- 31. Mississippian: Chappel, Barnett
- 32. Pennsylvanian: Caddo, Canyon, Bend Conglomerate, Flippen, Gardner, Harry Key Limestone, Hodge Edge [?], Hoxie and Moran

The Barnett Shale is an unconventional reservoir with low porosity and very low permeability. Porosity in productive siliceous shale intervals averages 5 to 6%, with permeabilities measured in the [inserted: 0.1 to 1] nanodarcy range. For comparison, tight sand reservoirs have permeabilities of .01 to .1 millidarcies (equal to 10,000 to 100,000 nanodarcy). To put it another way, the permeability of a tight sand is 4 to 5 orders of magnitude higher than the Barnett Shale. [struck: "a tight gas sand has 4 to 5 orders [higher]"]

According to Hill, Jarvie et al. (2007), approximately ³²⁴ [inserted: 3+] Bcf per section has been expelled from the Barnett Shale, either in the form of gas or oil equivalent. This indicates an expulsion efficiency of about 60%. [marg: "It is highly unlikely that any expulsion has taken place under the current conditions of nanodarcy permeabilities."] They also state that a net porosity increase of 4.3 to 4.6% results from the conversion of kerogen to hydrocarbons. This provides a considerable part of the known porosity of the

Barnett
Shale

Barnett Shale and results [marg: "(as a consequence)"] in a multiplicity of compartmentalized microreservoirs. Subsequently, to achieve gas flow in Barnett Shale wells, rigorous stimulation is required to rupture these microreservoir compartments.

Miocene Age — Monterey Shale (p.7)

[marg: "I) A) 12)"]

The Miocene-Monterey Shale, long known as the critical source rock in California, also includes significant fractured chert & porous diatomite reservoirs (5). This formation has been estimated to contain more than 500 billion barrels of original oil-in-place (1).

Production from the Monterey Shale actually began in the Santa Maria Basin in 1900. The Monterey Formation is probably the principal source rock in the Santa Maria Basin as there are no other rocks with good source characteristics in the basin (4).

The Monterey Shale's potential as a source rock has outshone its potential as a reservoir rock for decades throughout the state of California, as companies began developing such mammoth fields as Midway-Sunset, South Belridge, Kern River, Cymric and Elk Hills. It is estimated that 38 billion barrels of oil has been produced from fields relying on the Monterey as source rock (2).

II) Migrated Hydrocarbons That Came From Tight Source Rocks — That Still Contain Large Volumes of HC's (now are low-k reservoir rocks) (p.8)

- 1 33. Barnett sourced •
- 2 34. Brown Dense sourced the Smackover •
- 3 35. Eagleford sourced the Austin Chalk
- 4 36. Mancos sourced the Gallup Sandstone
- 5 37. Monterey sourced most of California •
- 6 38. Sunniland sourced •
- 7 39. Tuscaloosa sourced •
- 8 40. Utica-Pt. Pleasant sourced •
- 9 41. Woodford sourced
- 10 42. Mississippi Lime •

III) Major Sand Injectites & Re-Mobilization (p.9–10)

A) Associated With Hydrocarbons ^{A)}

- 1 43. Large Accumulations of Hydrocarbons ^{B)} North Sea: Alba, Balder, Cecile, Gryphon, Harding Prospect, Jotun, & Sleipner, Midland Valley, Scotland — Oil Shale Group (Dinantian)
- 2 44. Early Oil Formation & Migration Associated With Sand Injectites — example VIII) A) 1: Phillips (1990), Thompsons & others (1999) [?]

B) Large Sand Injectites Not Directly Associated With Oil and Gas

- 1 45. Malloy Slope — Norwegian Margin
- 2 46. Harcela [?] Formation — East Greenland
- 3 47. Miocene — Santa Cruz, California
- 4 48. Vicontian [?] Aptian — SE France
- 5 49. Early Paleocene — Panoche & Jumay [?], Central California
- 6 50. Cero Toro — Magallanes Basin, So. Chile
- 7 51. Jesus Formation — Marathon Basin, TX
- 8 52. Jurassic Eolian — Colorado Plateau

C) Mud Volcanoes

- 1 53. Carboniferous Ross Formation — Clare City [?], W. Ireland

IV) Long Distance Lateral Migration (p.11)

[marg, connected by arrow to top of list: "G) Alberta Basin"]

- A) Illinois Basin:
- B) Denver Basin:
- C) East Texas Basin:
- D) Louisiana–Wilcox Trend:
- E) Wyoming/Idaho [?] (Phosphoria) — to Big Horn Basin, Wind River Basin & possibly Powder River Basin
- F) Williston Basin — all the way to Canada
- G) Alberta Basin ^{*(see marginal note above)*}

V) Large Vertical Migration From Deep Sources (p.11)

- A) Los Angeles Basin:
- B) SW Alabama: Jurassic Smackover/BD to Upper Cretaceous Eutaw
- C) NE Louisiana: Jurassic Smackover/BD to Upper Cretaceous Monroe Bioherm [?]

- D) East Texas Basin:

VI) Hydrocarbons Sourced By Too Ancient of Source Rocks (p.12)

- A) Proterozoic

1 54. Beetaloo Basin — Northern Territory, Australia

- B) Pre-Cambrian

VII) Evidences of Early Oil Migration (p.13–16)

A 15 Diagenetic Traps (φ/k decrease below o/w contacts)

Each entry: *location (age/rock) — original citation era WYYYY → author(s), year.*
W1975/W1977/W1990/W2005 appear to be the author's own running-bibliography tags
(editions of a reference work, not publication years of the cited studies).

- 1 55. Arkansas: Jurassic Reynolds oolite fields (W1975→Vestal 1950)
- 2 56. Iraq: many fields (W1975→Dunnington 1958, 1967; W1977→Dunnington 1960)
- 3 57. Surgut Dome, Russia (W1975→Zarapov & Prozorovich 1970)
- 4 58. Russian Platform (W1975→Mirapolskaya 1970)
- 5 59. Ekofisk, N. Sea chalk (W1977→Scholle 1976; W1990→Scholle 1977, Feazel 1985, D'Heur 1984)
- 6 60. Abu Dhabi: Lower Cretaceous Thamama Carb. (W1977→Clarke 1975)
- 7 61. Fahud field, Oman: Cretaceous Wasia LS (W1977→Tschopp 1967)
- 8 62. Thomasville, Rankin Cty Miss., Smackover (W1977→Wilson 1975)
- 9 63. PR Basin, Wyoming & Montana, Cretaceous SS (W1977→Webb 1974, Scholten 1959)
- 10 64. NW Oklahoma, Penn. Morrowan SS (W1977→Adams 1964)
- 11 65. North Sakhalin Island, USSR (W1977→Yukova [?] 1970)
- 12 66. Denver Basin, Permian Lyons SS (W1977→Levandowski 1973)
- 13 67. N. Argentina & SE Bolivia, Andean Foothills, Permian SS (W1977→Reed 1946, Weeks 1958)
- 14 68. Pleasant Valley field, San Joaquin Basin, Gatchell SS (W1990→Schneeflock 1998; W1977→Harding 1976, Weddle [?])
- 15 69. Nancy field, Clarke Cty Miss., Smackover (W1977→Hughes 1968, Vistal [?] 1950, Bishop 1991)
- 16 70. Dukhan, Qatar, Jurassic No. 3 (W1977→Qatar Pet. Co. Staff 1956)
- 17 71. Walker Creek, Smackover (W1990→Becher & Moore 1976)
- 18 72. Clinton Sands of Ohio (W1990→Heald 1940)
- 19 73. Cadotte SS, BC, Lower Cretaceous (W1990→Thomas & Miller 1980)

- 20 74. SW Germany, Dalum field, Wealdum reservoirs (W1990→Fuchtbauer & Goldschmidt 1964)
- 21 75. Ekofisk (W1990→D'Heur 1984, Feazel 1985, Scholle 1977)
- 22 76. Tartan field, N. Sea (W1990→Burley 1986)
- 23 77. East Coalinga & Gujarral Hills / Pleasant Valley, San Joaquin Basin, Gatchell SS, Eocene (W1990→Weddle 1951, Harding 1976, Schneeflock 1978, Chambers 1943, Church & Krammes 1957)
- 24 78. Spindle field, Denver Basin, Tertiary & Codell SS/Dakota (W2005→Porter & Weimer 1982)
- 25 79. Gifhorn trough, NW Germany (W2005→Fuchtbauer 1967, 1992)
- 26 80. Clinton Sd, Ohio (W2005→Heald 1940)
- 27 81. Chester & Aux Vases, Illinois Basin (W2005→Swann [?] & Narian 1960)
- 28 82. Cadotte SS, NE British Columbia (W2005→Thomas & Miller 1980)
- 29 83. Dukhan field, Qatar (W2005→Dunnington 1967)
- 30 84. Asab field, Abu Dhabi (W2005→Johnson & Budd 1975)
- 31 85. Sahil field, Abu Dhabi (W2005→Oswald 1995)
- 32 86. Yibal field, Oman (W2005→Litsey 1983)
- 33 87. Buckhorn Asphalt Quarry, Sulfur, Okla. (W2005→Squires 1978, Brand 1989)
- 34 88. NE Thompsonville field, Eocene Wilcox (W2005→Stricklin 1994)
- 35 89. Elgin & Franklin, Fulmar Sand, gas condensate (W2005→Hinton 1998)
- 36 90. Brent Sand Reservoirs, North Sea (W2005→Hancock & Taylor 1978, Kantorowicz 1990, 1991)
- 37 91. Piper field, North Sea (W2005→Maher 1980)
- 38 92. North Sea, widespread evidence (W2005→Coleman & Robinson 1994)
- 39 93. Deep gas, Cretaceous Tuscaloosa (36% ϕ @ 20,000 ft, 680 mD) (W2005→Bloch 2002, Thompson 1979)
- 40 94. Chalk fields, Danish North Sea (W2005→W1990)
- 41 95. West Ekofisk (W2005→Vanden Bork & Thomas 1980)
- 42 96. Reclose Muddy SS, PRB (W2005→Webb 1974, Berg 1976)
- 43 97. South Glenrock field, PRB (W2005→Curry & Curry 1954)
- 44 98. Third Creek & Lucky Ditch, Denver Basin (W2005→Reinhart & Davies 1978, Reisser & Blaucke 1989)
- 45 99. San Andres, NW Shelf Permian Basin (W2005→Keller 1992)
- 46 100. Russian Basins (W2005→Vinogradov 1979)
- 47 101. Villeperdue, Paris Basin (W2005→Duval 1992)
- 48 102. Yucal-Placer gas field, Eastern Venezuela (W2005→Reistoffer & Laurier 1987)
- 49 103. Hamada Basin, NW Libya (Devonian SS) (W2005→Shah 1993)
- 50 104. Hatdan [?] field, Danish N. Sea (W2005→World Oil 2001)

VII B) Compaction Inhibition from Early HC Migration (p.15)

- 1 105. Asab oilfield, Abu Dhabi (W1990→Johnson & Budd 1975, Mallinson & Sharp 1975)
- 2 106. Yibal oilfield, Oman (W1990→Litsey 1983, Dunnington 1967)
- 3 107. West Ekofisk, North Sea (middle TOR reservoir) (W1990→D'Heur 1984)

- 4 108. East Hod, North Sea (W1990→D'Heur 1984)
- 5 109. El Bundug, offshore Qatar (W1990→Bashbush 1983, El-Aouar & Rasool 1970)

VII C) Basal Jur. [?] Muts (p.16) **TAR MATS**

(heading uncertain — possibly "Basal Jurassic ____"; original reads "Basal Jur muts")

- 1 110. McLouth field, KS (W1990→Lee & Payne 1944)
- 2 111. Hawkins field, Texas (W1990→Wendlandt 1956)
- 3 112. Ghawar, Uthmaniyah area, Saudi Arabia (W1990→McKeown [?] 1979, Aramco 1959)
- 4 113. Fereidoon field, Iran (W1990→Kaueh [?] 1971)
- 5 114. Minagish field, Kuwait (W1990→El-Aouar & Rasool 1970)
- 6 115. El Bundug field, Qatar (W1990→Bashbush 1983)
- 7 116. North Coles Levee / South Coles Levee / Elk Hills, Miocene Stevens Sand (W1990→Biles & Ramseyer 1987, Davis 1952, Hardwin [?] 1962, Lorshbaugh 1967, Maher 1975, Peterson & Hickey 1984, Simonson 1959)

VIII D) Closed Systems — Isolated From Migration Paths (p.17)

- 1 117. Racoon Bend field, Texas, Austin Cty, Oligocene Gyroskey [?] Sands (W1975→Texas & Miller 1933, p.1477)
- 2 118. Milbur field, Texas, Eocene Wilcox barrier bar (W1975→Chubar 1972)
- 3 119. Thrall field, Texas, Cretaceous Taylor Marl serpentine plugs (W1975→Sollards 1932)
- 4 120. Near Clinc [?] / west of Uvalde, Texas, Anacacho LS lens (W1975→Adkins 1932)
- 5 121. Ecuador, olistoliths embedded in source rock (W1975→Coleman 1970)
- 6 122. Aneth Area, Paradox Basin, Utah, Penn. carbonates (W1975→Peterson 1966)
- 7 123. Bisti field, New Mexico, Late Cretaceous offshore bar sand lens (W1975→Sabins 1963)
- 8 124. Calhoun field, Louisiana, Cockfield clastic (W1975→Exum 1973)
- 9 125. Mid-Continent, Oklahoma & Kansas, Penn. Cherokee sand lenses (W1975→Busch 1959)
- 10 126. Tertiary Delta, Louisiana, closed-system clastic trap (W1975→Clark & Rouse 1971)
- 11 127. Denver Basin, Colorado, Cretaceous Dakota sand lens (W1975→Helberg 1964)
- 12 128. Maikop District, Russia, Tertiary sands (W2005→Prokopov & Maximov 1937)
- 13 129. Colgrade field, Louisiana, Eocene Wilcox (W2005→Nelson 1963)
- 14 130. Orinoco Basin, Venezuela, Oficina sand lenses (W2005→Banks 1959, 1966)
- 15 131. South Texas, Cherokee Group, Desmoinesian (W2005→Baker 1962, Banks & James 1974)
- 16 132. SW Mississippi, Wilcox sand lenses (W2005→Culbertson 1969)
- 17 133. Derna Basin, Romania, Pliocene sands (W2005→Sproul 1938, Walters 1974)

- 18 134. Chinese Neogene, immature oil in lenticular traps, lacustrine sediments
(W2005→Zhang Y.-Gang 1981)

VII E) Traps That Are Barren Only Because of Late Trap Formation (near to older, productive traps) (p.18)

- 1 135. Large Hatchetigbee Anticline, SW Alabama (barren) (W1975→Hopkins 1917, Keyes 1971, Oxley 1968, Dilcas J. 1972)
- 2 136. Kelsey Anticline (barren), Woodbine SS; Opelika (barren); Hawkins Dome (productive); East Texas field (productive) (W1990→Denison 1933, Wendlandt 1946)
- 3 137. Sabriyah field, barren Zubair Sands (Kuwait); Raudhatain field (productive) (W1990→Al-Rawi 1981, Adasoni 1967)
- 4 138. Adkins (1956) study of 200 East Texas salt domes proved that if a trap was not in place by the time 2,000 feet of overburden [had accumulated], then it was not productive (W1990→Adkins 1956)
- 5 139. Smackover Gulf Coast — if the trap was not developed by Lower Buckner time, it will not be productive (W1990)
- 6 140. San Felipe Dome, Hockley Dome (Eocene Wilcox & Yegua saltwater); Katy field (Eocene Wilcox & Yegua gas reservoir) (W2005→Halbouty 1979)
- 7 141. Qatar Arch (Late Jurassic, void of Jurassic HCs, even though the arch separates 2 oil kitchens) (W2005→Wilson 1982)
- 8 142. Destin Dome (very large structure, offshore Pensacola, Florida) — first well drilled in 1974 by Exxon; by 1977 had fifteen extremely costly dry holes (W1975b [Wilson]; W2005→Durham 2001)
- 9 143. Alaminos Canyon (high, late Bama structure, blocking Wilcox paleostructure); "112 mm dry hole" / Kessel [?] 26 of Wilcox (low-relief Wilcox, paleostructures have [1150' of HC column — Walker Ridge field]); Cascade field; Walker Ridge field (W2005→Meyer 2005)